Educated to Be Mothers? Indoctrination and Demographic Backlash*

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Abstract

Social norms play a critical role in shaping fertility choices. This paper shows that public policy aimed at increasing fertility by reinforcing traditional gender roles can backfire. We study the 1945 primary education reform implemented under Franco's regime in Spain, which promoted nationalist-religious values and emphasized women's domestic roles. Exploiting plausibly exogenous variation in exposure across birth cohorts ~~within~~ a difference-in-differences framework, we find that women exposed to the reform bear significantly fewer children ~~than their unexposed counterparts~~. We further provide suggestive evidence that exposed women are more likely to reject the regime's promoted gender norms and family ideals, pointing to the possibility of a cultural backlash. At the aggregate level, this individual-level decline translates into a marked reduction in fertility rates in areas with a larger pre-reform share of potentially affected women. Taken together, our findings highlight the unintended long-term consequences that state-led efforts to engineer social norms—particularly under autocratic regimes—can have on demographic behavior.

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1 Introduction

Over the past half-century, fertility rates have fallen dramatically worldwide. The global total fertility rate has declined from around 5 children per woman in 1960 to about 2.3 today, and an increasing number of countries are now below the replacement level of 2.1 births per woman (Rindfuss et al., 2010; Hart et al., 2024). This decline has raised concerns about population aging, shrinking workforces, and associated economic challenges (Hart et al., 2024). In response, many governments, especially in developed nations and a growing number of developing countries, have adopted pro-natalist policies aimed at encouraging childbearing. According to United Nations data, by the mid-2010s about 66% of European countries and 40% of Asian countries had policies to raise fertility or arrest its decline (Hart et al., 2024). These policies take various forms, including direct financial incentives (baby bonuses, child allowances, tax credits), expanded parental leave, subsidized childcare, housing benefits for families, and other measures to reduce the cost of child-bearing.

While the economic cost of parenthood plays a central role in shaping fertility behavior, demographic trends may also be affected by the social and cultural context in which family-formation decisions are made. Cultural values and gender norms governing the division of domestic labour have been shown to influence the time and opportunity costs of child-rearing and, in turn, fertility behaviour (Doepke and Kindermann, 2019; Sakamoto and Minami, 2024). While individuals from larger families tend to hold and transmit traditional family values across generations (Vogl, 2016), comparative analyses show that countries where women participate extensively in the labour market but retain a disproportionate share of domestic work tend to maintain below-replacement fertility despite substantial family benefits (Esping-Andersen and Billari, 2015; Goldscheider et al., 2015). These patterns have also featured prominently in policy discussions, where some policymakers have linked low fertility to changes in gender roles, family formation norms, and work–life preferences. In the United Kingdom, for example, public figures have called for broad cultural changes to encourage childbearing (Farage, 2024), while in the United States, pro-natalist arguments have emphasised the potential demographic effects of a return to more traditional family structures (Barkhorn, 2025). From this perspective, restoring traditional gender roles and family structures should be beneficial for demographic recovery.

This paper provides novel evidence that public policy aimed to stimulate fertility by stressing traditional gender roles can have unintended effects on fertility decisions by generating a backlash among treated women. We show this by focusing on the indoctrination carried out by the regime of general Francisco Franco right after the victory of the Nationalist band in the Civil War that took place in Spain from 1936 to 1939. Franco ruled over Spain from 1939 to 1975 as a dictator. The first primary education law during the Francoist rule was approved in 1945 and it is the subject of empirical study in this paper. The law outlined the key features of primary education,

emphasizing religious instruction, strong national spirit, mandatory use of the Spanish language, and a specialized curriculum for girls focused on home life, crafts, and domestic industries. The curriculum was modified in all schools throughout the country to teach traditional gender roles to children enrolled in private and public schools. The regime established strict guidelines for methodological practices, carefully regulating time management and the structure of the school day. Furthermore, the state also mandated the separation of sexes (compared to mixed schools that were in place during the Second Republic) in primary education, while kindergarten schools and co-educational schools would always be run by female teachers.[1]

A key goal of the reform was to educate women to become "good wives" and "good mothers", as that would constitute the most appropriate way for them to contribute to society (Toboso Sánchez, 2009, p. 213). As such, this initiative was not an isolated case. Historical examples of autocratic regimes trying to interfere in the private lives and fertility decisions of their citizens through propaganda and/or indoctrination include the Lebensborn program created by the Nazi authorities to encourage fertility among German women deemed "racially valuable" (US Holocaust Memorial Museum), the award of Medals of motherhoods in the Soviet Union (Levin) and the National Birthrate Awards to promote an increase in the country's fertility rate established by the Franco regime in Spain.

To uncover the causal effect of Francoist indoctrination on fertility behavior, we conduct an individual-level analysis that uses both nationally representative survey data and census microdata in a difference-in-differences framework. Our identification strategy leverages quasi-exogenous variation in exposure to Franco's 1945 primary school reform, stemming from the interaction between individuals' birth cohorts, the implementation date of the reform, and the official age of entry and exit from primary education. Our intention-to-treat (ITT) variable measures the share of a respondent's primary schooling years that took place under the post-reform curriculum. Using this measure, we find that women more intensively exposed to the reform—due solely to the timing of their school entry—are equally likely to become mothers, but they have significantly fewer children on average. This suggests that the reform influenced fertility decisions primarily along the intensive margin. To further validate these findings, we replicate the analysis on a different and much larger sample, namely the Spanish census microdata. We also rule out that the results are driven by cohort-specific effects common to other Mediterranean autocracies of the period by running a placebo test using ESS data for Portugal.

Additional analysis provides no support for a set of hypothesised mechanisms, including differen-

[1] Co-educational schools would not be authorized except in particular cases where the population would not provide a school contingent of more than 30 students between the ages of 6 and 12. Primary school comprised four phases: nursery schools for children aged four or younger, kindergarten (4-6 years old), which could admit both boys and girls when enrolment would not allow for gender division, elementary education (6-10 years old), improvement (10-12 years old), and vocational initiation (12-15 years old).

tial labour market participation, educational achievement, and political leaning. In contrast, our analysis provides at least strongly suggestive evidence that the fertility decline among treated cohorts was accompanied by a shift in beliefs plausibly shaped by the reform—particularly regarding gender roles, motherhood, and the family. These results are consistent with the presence of a backlash effect, whereby the regime's attempt to engineer traditional values ultimately undermined the very same norms it sought to entrench.

Data limitations do not allow us to quantify the extent to which this backlash was caused by mere exposure to the reform versus other factors, such as the different curricula, the division of classes, or the different school material between males and females. However, to investigate whether these individual-level effects aggregated into broader demographic shifts, the final part of the paper turns to historical census data and it shows a second set of difference-in-differences specifications at the cohort–locality level. Here our ITT is defined as the proportion of the female population of primary school age in each locality at the start of the policy. In this setting, we compare fertility trends across areas that differed in the share of school-age girls at the onset of the reform—a demographic factor that plausibly shaped the intensity of exposure to Francoist educational indoctrination across localities, and subsequently influenced fertility outcomes as these cohorts reached reproductive age. This analysis reveals a significant decline in fertility rates in areas with greater ITT exposure to the reform, while marriage rates remained unaffected. A variety of robustness checks confirm this conclusion and placebo tests using the share of school-age boys yield null effects, mitigating concerns that our estimates might merely capture pre-existing demographic trends unrelated to the reform.

Our paper contributes to four different strands of literature. First, our findings contribute to a large body of literature on family policies and fertility. The main three policy areas involve transfers, parental leave, and childcare support. Direct financial incentives to parents include one-time "baby bonuses" for each birth, recurring child allowances, or tax credits/deductions for families with children. Financial incentive policies effectively lower the monetary cost of an additional child and, although the magnitude of the effect varies, they generally show a positive causal impact on fertility. Examples include Quebec's "Allowance for Newborn Children" programme (Milligan, 2005), Israel's child allowance system reform (Cohen et al., 2013), Russia's maternity capital policy (Slonimczyk and Yurko, 2014), Spain's baby bonus (González, 2013; González and Trommlerová, 2023), Singapore's baby bonus (Malak et al., 2019), Romania's fixed benefits reform (Tudor, 2020), and South Korea's allowances (Kim et al., 2023).

A major set of pro-natalist interventions are also parental leave policies that guarantee job-protected leave from work for mothers or fathers after a child is born, often with some wage replacement. Over the last decades, many countries have extended the duration of maternity leave, introduced paid leave, or added paternity leave for fathers. The empirical literature finds mixed but generally positive effects of such leave reforms on fertility in Austria, Norway and the

U.S. (Lalive and Zweimüller, 2009; Dahl et al., 2016; Cannonier, 2014; Golightly and Meyerhofer, 2022; Bailey et al., 2025), with clearer fertility impacts linked to more substantial leave reforms or to implementations in contexts with previously very limited support (see, e.g., Raute, 2019 on the effects of the 2007 parental leave reform in Germany). Some recent studies also find that giving fathers parental leave (paternity leave quotas or shareable leave) causes little change in fertility (Lee et al., 2009; Cools et al., 2015; Farré and González, 2019; Duvander et al., 2020).

Some studies also investigate the impact of childcare support policies on fertility (Rindfuss et al., 2010; Havnes and Mogstad, 2011; Mörk et al., 2013; Bauernschuster et al., 2016; Wood and Neels, 2019). The general takeaway from these studies is that childcare support appears one of the most effective pro-natalist policies in the empirical literature, as evidence from Norway, Sweden, Germany, and other countries consistently shows positive fertility effects when childcare becomes more available and affordable.

Beyond the main three policy areas (i.e., transfers, parental leave, and childcare support), various other measures have been tried or proposed to encourage childbearing. The economic literature on these is sparser but growing. A few notable areas include housing policy (Clark, 2012; Fazio et al., 2024), tax incentives (Parent and Wang, 2007; Cohen et al., 2013; Laroque and Salanié, 2014), health and family planning (Abramowitz, 2018; Gartner et al., 2022), and broader socio-cultural campaigns (Zeng and Hesketh, 2016; Lin et al., 2024). However, no studies to date have empirically examined whether propaganda and indoctrination may work to stimulate fertility in future generations. Our results rectify this omission and provide no empirical support for the claim that indoctrination in school stimulates fertility - rather the opposite, as they show that efforts to exacerbate gender stereotypes are likely to backfire and result in reduced fertility (Botev, 2015). This conclusion is of primary importance to a policy maker interested in combating the modern demographic winter.

Second, this paper relates to the existing research on gender norms and how they affect the lives of women in many dimensions (Bertrand et al., 2015; Fortin, 2015; Goussé et al., 2017; Andresen and Nix, 2019; Grewenig et al., 2020; Olivetti et al., 2020; Cortés and Pan, 2023; Boelmann et al., 2025). Gender norms play a crucial role in defining gender inequality (Kleven, Landais, Posch, et al., 2019; Bertrand, 2020; Ashraf et al., 2022; Bursztyn et al., 2023; Farré, Felfe, et al., 2023), and modern gaps can be traced at least in part to parenthood, which has the potential to hold back women's careers relative to men (Daniel et al., 2013; Angelov et al., 2016; Kleven, Landais, and Søgaard, 2019; Bertrand, 2020; Cortés and Pan, 2023). Traditional norms typically assign to women the role of main caregivers within the family and restrict their options in the labour market after childbirth (Bertrand et al., 2015; Fortin, 2015; Goussé et al., 2017; Andresen and Nix, 2019; Grewenig et al., 2020; Olivetti et al., 2020; Cortés and Pan, 2023; Boelmann et al., 2025). While children inherit gender norms from their parents (Fernández et al., 2004; Alesina, Giuliano, et al., 2013; Farré and Vella, 2013), thus possibly perpetuating

gender imbalances at home and making policy ineffective in reducing gender differentials in the labour market (Albanesi et al., 2023; Kleven, Landais, Posch, et al., 2024), public policy aimed at promoting gender equality in the long term has the potential to effectively shape the gender norms of the next generation (Hara and Rodriguez-Planas, 2021; Farré, Felfe, et al., 2023). Our paper contributes to this debate by showing that policies in the opposite direction, i.e., trying to stress traditional gender roles and gender stereotypes, are unlikely to achieve their intended purpose and may reduce fertility. This finding echoes Doepke and Kindermann (2019)'s prediction that, in low-fertility settings where it is often women who veto having additional children due to the disproportionate cost they bear in terms of career and time, fertility should be highly responsive to targeted policies that alter the childcare burden specifically for mothers and the feasibility for women to combine work and family.

Third, our study relates to the existing research on the impact of indoctrination and propaganda by authoritarian regimes (Waldinger, 2010; Waldinger, 2012; Adena et al., 2015; Voigtländer and Voth, 2015; Fuchs-Schündeln and Masella, 2016; Cantoni et al., 2017; Homola et al., 2020). In authoritarian regimes as in modern democracies, the content and quality of education exerted lasting consequences for people's beliefs. Germans who grew up under Nazis had access to a significantly lower quality of education (Waldinger, 2010; Waldinger, 2012) and they are much more anti-Semitic than those born before or after that period (Voigtländer and Voth, 2015). East Germans who were exposed to communism are more in favor of state intervention than West Germans (Alesina and Fuchs-Schündeln, 2007) and prolonged exposure to socialist education reduced their likelihood to obtain a college degree (Fuchs-Schündeln and Masella, 2016). In Spain, Catalan teaching was effective in increasing Catalan feelings (Clots-Figueras and Masella, 2013), while in China modifying the school curriculum was effective in changing young people's views of China's governance and free trade (Cantoni et al., 2017). On the other hand, language restrictions in elementary schools in the U.S. after World War I instigated a backlash, increasing the sense of cultural identity among the minority, reducing integration and identification with the host country later in life, and showing that indoctrination and propaganda may not always generate the intended effects (Fouka, 2020). Although these studies convincingly document the lasting effects of the content and quality of education on people's beliefs, to our knowledge this is the first paper to analyse the impact of indoctrination in school and authoritarian propaganda on fertility decisions.[2]

Finally, our analysis contributes to the existing research on the formation of cultural beliefs, the marriage market and fertility (Chong and La Ferrara, 2009; La Ferrara et al., 2012). While existing studies show that *novelas*, soap operas and TV series have the potential to affect the marital and fertility decisions of individuals, our analysis highlights that indoctrination and propaganda

[2]At the time of writing, we are aware of one unpublished manuscript studying the analysis of Nazi family policies on fertility by Baudin and Stelter (2023).

are ineffective policy tools to affect the marriage market and they are counterproductive to spur fertility and demographic growth.

The rest of the paper is structured as follows. Section 2 revises the historical background of Spain. Section 3 shows the data sources that form our dataset. Section 4 describes the empirical analysis and Section 5 presents the results. Finally, Section 6 concludes.

2 Historical background

In 1936, a military uprising led by Francisco Franco aimed at overthrowing the Republican government unleashed the Spanish Civil War. The conflict between Republicans and Nationalists, which ended in 1939 with Franco's victory, resulted in approximately 500,000 fatalities and around 450,000 people exiled (Castro Oury, 1993; Preston, 2011, p. 608).

The Francoist regime repeatedly expressed its determination to implement demographic policies, considering fertility one of the key concerns of the state (Nash, 1991, p. 160). Ramón Serrano Suñer-president of *FET y de las Juntas de Ofensiva Nacional-Sindicalista, JONS*- pointed out the necessity of a *"serious demographic policy to accomplish the ideas of the Falangist State"*. In the words of Ramiro Ledesma, founder of the political party *JONS*: *"we can only seriously speak of Spain's superiority when it becomes attainable through a population that has at least doubled" (p. 33 Bosch Marín, 1941)*. Indeed, maternity was described as an *"ordinary way of life to reach perfection"* (Martínez Medrano, 1959, p. 31) and represented the feminine goal: *"the feminine profession should be oriented towards family life: it belongs to the very essence of a woman to be always and everywhere a mother, to devote and sacrifice herself, and to care attentively for others"* (Jaén Sánchez, 1959, p. 21).

As part of Franco's agenda to boost fertility rates in post-civil war Spain, the anti-abortion and anti-contraception law was soon approved in 1941 (Spanish Official Bulletin, 1941, p. 768-770). In 1945, the first Primary Education Law of the Francoist regime was also enacted. The reform established that the specific education of women as wives, mothers, and educators of their children should begin in primary school (Rabazas Romero and Ramos Zamora, 2006, p. 46 and Szczygieł, 2014, p. 295-296). The reform was openly endorsed by the primary education inspector, Francisca Bohigas, who claimed that *"the girl was born to stay at home"* (p. 10, Bohigas, 1941), and it was backed by the Catholic Church and the Women's Section of the Falange (Toboso Sánchez, 2009, p. 212).

Primary education was divided into three periods: elementary (ages 6-10), advanced (ages 10-12) and introduction to professional practice (ages 13-15). The law defined the main characteristics of primary education: a different curriculum for girls preparing them for home life, crafts, and domestic industries, among others as they must be prepared to become mothers and wives

(Rabazas Romero and Ramos Zamora, 2006, p. 47; Francés Díez, 2013, p. 226). Moreover, the law mandated the inclusion of religious education, emphasized the promotion of strong national identity, and enforced the use of the Spanish language. The role of teachers was subordinated to the supreme interests of the nation, with the responsibility of contributing to the formation of girls as future mothers.

The Female Section contributed to the regime's propaganda efforts through the journal *Consigna*, published from 1940 to 1977 and containing pedagogical guidelines for female teachers, mothers, or future mothers (p. 49–50 Rabazas Romero and Ramos Zamora, 2006):[3]

> *"From girlhood, a woman is feminine, and her differentiation from boys is evident even in the smallest details and games. The care invested in her education will later be reflected in the home, where the girl, now a woman, must shape all those around her: husband, children [...] And isn't it a serious responsibility of a teacher—or of an educational system—not to have fully nurtured the seed that was planted? [...] If we enable our women to become complete women, we will have eliminated many of the unhappy and uncomfortable households that exist in our society today. Because the mother woman will raise her daughters just as we have raised her [...]. This is a call to the conscience and responsibility of every female teacher." (p. 30–32, Consigna, 1958)*

The regime issued precise guidelines on methodological practices and strictly regulated time usage during the school day.[4] Additionally, it mandated the separation of sexes in primary education and kindergarten and coeducational schools to be run by female teachers.The law also established the compulsory subjects to be taught in school: core subjects (such as writing and arithmetic), formative subjects (such as mathematics), and complementary subjects (such as feminine tasks, including domestic economy and family life education; see Toboso Sánchez, 2009, p. 215). Moreover, some schools organized professional initiation courses for children aged 12-15 years old.[5] School material, such as books, had to be approved by the Ministry of National Education and the ecclesiastical hierarchy (González Pérez, 2009, p. 97).[6]

Numerous examples of the different reading books used to educate boys and girls during the Francoist regime document the indoctrination perpetrated by the regime. The latter were given books that narrated the birth of the protagonists (though without referencing sex, instead

[3]The Female Section (*Sección Femenina*) was founded in 1934 and led by Pilar Primo de Rivera, sister of José Antonio Primo de Rivera, founder of Falange. In 1937, a Social Service, organized by *Sección Femenina*, was required for women aged 17-35, aimed at those who aspired to occupy positions in the administration or obtain university degrees (Gobierno del Estado, 1937). We choose not to analyze this gender-specific reform due to potential endogeneity concerns arising from selection bias.

[4]School attendance was compulsory from 6 to 12. Penalties were imposed on those who required children to work. For the first time, a school record book and a primary education certificate were established as documents attesting to the compulsory nature of education.

[5]In particular, the law differentiated "domestic-type female facilities where students, according to their gender, would carry out appropriate practices to develop work habits, prepare for professional life, and their future family roles."

[6]See the Spanish Official Bulletin (1945, p. 385-394).

attributing it to God or the stork), stories where young girls were responsible for entire families, or tales emphasizing the important role of women as mothers (Mahamud, 2016), as seen in this fragment from the book titled *"Read me, girls"* (see also Figure 1):

> *"Little girls, budding women:*
> *I offer you one teaching: May you always feel in your soul the sacred fear of God, may you always be sweet little mothers [...], eyes to ease pain, little hands to heal suffering. [...] you will always be the sweetest, the most beautiful, the most delicate in the world." (Torres, 1943, p. 158)*

Figure 1: Cover and excerpt from *Leedme, niñas. Segunda parte* (Torres, 1943).

The volume *"Guirnaldas de la Historia"*, intended for girls, is another example that describes procreation as the ultimate goal to contribute to the nation (Criado and De los Reyes Leoz, 2019, page 292). This vision of womanhood was consistently reinforced in other materials too, such as:

> *"We can prepare as good mothers those that are still girls" (p. 7–11, Paya Ibars, 1959)*

> *"The boy will set his eyes on the world; the girl, on the home" (p. 215, Toboso Sánchez, 2009)*

The historical evidence presented here is of course purely anecdotal, but it exemplifies how

interfering in women's fertility decisions was a central objective of the 1945 Francoist primary education reform.

3 Data

In order to examine the impact of the 1945 Francoist primary education reform on fertility at the individual level, the empirical analysis draws on both the Centro de Investigaciones Sociológicas (CIS) barometer and specific thematic surveys as the primary data source. The CIS surveys are nationally representative surveys administered at irregular intervals since 1979. In addition to collecting standard socio-demographic information, these surveys elicit respondents' attitudes, values, and beliefs on a broad range of topics—including politics, religion, gender roles within the family and society, and views on childrearing. From the 1979 and 1990–1992 waves, we extract information on the total number of children ever born to each female respondent, which we use to construct our main individual-level outcome variables—both at the extensive and intensive margins. We also obtain information on each female respondent's year of birth, which allows us to assign exposure to Franco's school reform, and location of residence at the time of the survey.[7] From the 1979 and 1986–1988 waves, we further collect data on the labor force participation of women (i.e., whether women are employed, unemployed or housewives), their educational attainment, and whether they voted for right-wing parties in the last election.[8] To explore potential mechanisms—particularly the hypothesis of a cultural backlash triggered by Franco's school reform—we turn to the 1990 survey, which includes several questions probing female respondents' beliefs about women's role in the family, motherhood, female labor market participation, and alignment with Catholic values.[9]

To further validate the findings obtained from the CIS surveys, we complement our analysis with two additional sources of individual-level data: the Spanish census microdata produced by INE, and the European Social Survey (ESS). Regarding the former, census data—by virtue of their administrative nature and substantially larger sample sizes—allow us to improve the precision of our estimation relative to those derived from the comparatively smaller CIS samples. However, this advantage comes at the cost of limited information: census records lack the rich attitudinal content of the CIS surveys, particularly concerning the childbearing information for men, as well as beliefs, values, and social norms. As such, our analysis based on census data is limited

[7]For matter of privacy, CIS provides the exact municipality of residence only for localities with more than 100,000 inhabitants; thus, we classify localities into provincial capitals versus the rest of the province.

[8]Information on voting behavior is available only for the years 1986-1988.

[9]More specifically, we use the answer to questions such as: *Of the following types of families, which one best matches your ideal family? To what extent do you think a man should share household chores? Is sharing household chores very important to you? Is having children very important for achieving the stability of the couple? Do you believe that being a mother is an experience one must have in order to truly feel like a woman?* Appendix Table A2 shows the complete list of questions.

to testing whether our main result—the effect of exposure to Franco's primary school reform on women' fertility behavior—can be replicated in administrative data. Moreover, drawing on the ESS, we construct a cross-country placebo test using a state comparable to Spain in several key aspects but that was not exposed to the Francoist reform, namely Portugal.[10] This exercise allows us to assess whether the pattern observed in Spain may be driven by a broader cohort effect rather than the reform itself.

For both the Spanish census microdata and the ESS, we construct our dependent variables following the same logic used in the CIS-based analysis. From the census we use the 1991 and 2011 waves,[11] while from the ESS we use the 2006 and 2018 waves.[12] As in the barometer data, we collect information on respondents' gender and municipality of residence at the time of the survey,[13] and construct our main explanatory variable—exposure to Franco's school reform—based on the respondent's year of birth, which is consistently reported across both data sources.

To complement our baseline individual-level analysis and assess whether the effects of Franco's primary school reform are also detectable at the aggregate level, we compile a panel dataset spanning nearly five decades (1930–1979). The units of observation are provincial capitals and the remainder of their respective provinces, based on data availability across sources. Our primary outcome variable is a measure of fertility constructed from the annual Natural Movement of the Population reports published by INE. These reports provide disaggregated birth statistics for each provincial capital and for the rest of the province, allowing for spatial and temporal comparisons over the long run.

Ideally, aggregate fertility would be best captured by the Total Fertility Rate (TFR)—a weighted average of age-specific fertility rates for women of childbearing age (15–44). The TFR allows for meaningful comparisons both within and across observational units. However, the Natural Movement data report only the absolute number of live births per unit and year, without any disaggregation by maternal age. To make this crude birth count comparable across time and space, we adjust for the size of the relevant at-risk population. Specifically, we control for the number of women of reproductive age (defined as ages 15–44) in each observational unit and decade, as reported in the decennial Historical Censuses published by INE between 1930 and 1970.[14]

[10] Portugal's geographic proximity, strong Catholic tradition, and the presence of an authoritarian regime during the same historical period make Portugal an ideal comparison state for this analysis.

[11] The 1991 census is the earliest wave for which individual-level microdata are publicly available. We exclude the 2001 census since it does not contain fertility information.

[12] These are the only waves containing information about childbearing decisions.

[13] Unfortunately, municipality of residence is not available in the ESS.

[14] To account for potential shifts in the age structure of the reproductive population, we further disaggregate this group into three subcategories—ages 15–24, 25–34, and 35–44—and include separate controls for each

From the same INE Historical Censuses, we also collect data on the size of the female population aged 0–14 in 1940. This cohort serves as our proxy for exposure to Franco's school reform: being the first to be directly affected by the policy when it came into force in 1945, we expect its relative size in each observational unit to be predictive of the potential impact of the reform on future fertility patterns at the unit level as this cohort enters the reproductive age in the following years. Figure B1 displays the spatial distribution of our aggregated measure of treatment exposure. As shown, the intensity of exposure is relatively homogeneous across Spain in terms of North–South variation. However, some heterogeneity emerges along the coastal–inland axis: inland regions tend to exhibit higher levels of exposure compared to coastal areas. This spatial pattern provides reassurance that our empirical strategy is not confounded by broad North–South gradients, although finer geographic variation may still play a role in shaping the estimated effects.[15] In addition, we extract two further variables to support our analysis: also from the Natural Movement reports, we gather annual data on the number of marriages per observational unit; and from the historical censuses, we collect the size of the male population aged 0–14 in 1940, as well as the size of female and male population aged 15-24 in the same year.

Appendix A2 provides a list of the variables included in the channel analysis and brief descriptions, while Appendix Table A displays summary statistics for the main variables.

4 Identification

To estimate the causal effect of the Francoist primary school reform introduced in 1945, our baseline analysis exploits variation in exposure driven by the interaction between individuals' year of birth, the timing of the reform, and the official school entry and exit ages at the time.[16] In particular, we compare fertility choices across three groups of individuals: (i) those born between 1920 and 1929, who completed their entire primary education before the reform; (ii) those born between 1930 and 1939, who were partially exposed to the reformed curriculum; and (iii) those born between 1940 and 1950, who completed their full primary education under the post-reform

subgroup in our regressions.

[15]We account for this heterogeneity through the inclusion of fixed effects at the unit level in our regression models.

[16]As detailed in Section 2, primary school in Spain typically spanned ages 6 to 15. Prior to the Francoist reform, compulsory schooling roughly corresponded to the age range 6–14 (Instrucción Pública, 1923). After the reform, compulsory education formally ended at age 12, though students could continue attending primary school up to age 15 on a voluntary basis. These additional years were often attended by students who had repeated at least one grade—a relatively common occurrence at the time, particularly in rural areas where illiteracy was widespread. Others who remained enrolled beyond age 12 typically had not passed the entrance exam for upper secondary school (*bachillerato*), lacked access to vocational education, or stayed on to receive complementary instruction. To ensure consistency in the definition of primary schooling across the pre- and post-reform periods—and to remain agnostic about the motivations for voluntary continuation—we define primary education as spanning ages 6 to 15, and assign exposure accordingly.

system.[17] Specifically, we estimate the following difference-in-differences specification using the sample of women interviewed in the CIS barometer surveys:

$$y_{it} = \beta_0 + \beta_1 \mathit{1930/1939}_i + \beta_2 \mathit{1940/1950}_i + Year_t + Residence_i + AgeClass_i + \epsilon_{it} \tag{1}$$

where the dependent variable is either a dummy equal to 1 if woman i has ever had at least one child by year t, and 0 otherwise, or the total number of children she has had up to the year of the survey. The dummy variable *1930/1939* equals 1 if the respondent was born between 1930 and 1939; the associated coefficient β_1 captures the effect of partial exposure to the reform on fertility decisions, relative to the omitted category of women born between 1920 and 1929. Similarly, the dummy variable *1940/1950* equals 1 for women born between 1940 and 1950; the coefficient β_2 estimates the effect of full exposure to the reform on fertility outcomes.

The use of multiple survey waves allows us to disentangle the effect of the reform—which is inherently cohort-based—from any time-specific shocks by including year fixed effects ($year_t$). Although we cannot include fixed effects for respondents' exact age, we group individuals into ten-year age brackets at the time of the survey and include fixed effects for these age groups in the regression ($AgeClass_i$).[18] We also control for $Residence_i$, a set of fixed effects capturing the respondent's place of residence at the time of the interview, classified as either a provincial capital or the rest of the province. In our most comprehensive specification, we further allow for non-parametric trends that vary jointly by age class and place of residence by including the full set of interactions between the three groups of fixed effects. Standard errors are heteroskedasticity-robust and clustered at the birth cohort-survey year level.[19]

To test the validity of the parallel trends assumption and to gain further insight into the dynamic effects of the treatment in the years following the reform, we also estimate an event study specification, as shown in Equation 2 below:

$$y_{it} = \alpha + \sum_{\substack{c=1920 \\ c\neq 1929}}^{1950} \beta_c \cdot \mathbf{1}\{\text{BirthYear}_i = c\} + Year_t + Residence_i + AgeClass_i + \epsilon_{it} \tag{2}$$

In this setup, rather than grouping individuals into broader birth cohorts as described above, we include a set of indicator variables for each single-year birth cohort. The reference category is the last cohort to have completed the entirety of primary school before the Francoist reform,

[17] To maximize comparability, we restrict the last group to individuals born in 1950.

[18] Specifically, we define the following age groups: under 20, 20–29, 30–39, and so forth.

[19] The reason for this is twofold. First, clustering only by birth cohort—the level at which treatment is assigned—would result in a small number of clusters (only 31), which may lead to underestimated standard errors. Second, clustering by both birth cohort and survey year better accounts for residual correlation among respondents who share both exposure timing and survey wave.

namely those born in 1929.

5 Results

We begin the discussion of our results by presenting in Table 1 the baseline estimates of the impact of the Primary Education Reform on women's childbearing decisions. Columns 1–4 report estimates using a binary outcome indicating whether a woman has ever had a child (extensive margin), while Columns 5–8 present estimates for the total number of children ever had (both extensive and intensive margin). Column 1 and 5 include only year fixed effects; Columns 2 and 6 add fixed effects for place of residence; Columns 3 and 7 additionally include 10-year age group fixed effects; Columns 4 and 8 incorporate the full set of interactions among these three groups of fixed effects. As shown in Table 1, we find no statistically significant effect of the reform at the extensive margin: women who were partially or fully exposed to the reform appear equally likely to have ever had a child compared to the control group. However, when we consider both the extensive and intensive margin, we find a statistically significant and negative effect limited to the fully treated cohorts. In particular, women who completed their entire primary education under the reform exhibit a lower average number of children compared to untreated cohorts. In the most saturated specification (Column 8), the reform appears associated with a reduction of approximately 0.34 children—equivalent to a 12% decrease relative to the average number of children among untreated women. In other words, the reform did not significantly influence the decision to become a mother, but it did affect family size—that is, the number of children women chose to have.

Table 1: The impact of Francoist educational reform on childbearing decision: DiD

	(1) Kids	(2) Kids	(3) Kids	(4) Kids	(5) N. kids	(6) N. kids	(7) N. kids	(8) N. kids
1930/1939	0.012	0.010	-0.014	-0.006	-0.079	-0.061	-0.056	-0.177
	(0.013)	(0.013)	(0.020)	(0.019)	(0.069)	(0.067)	(0.106)	(0.131)
1940/1950	0.017	0.013	-0.004	0.002	-0.370***	-0.401***	-0.260**	-0.344**
	(0.010)	(0.009)	(0.025)	(0.025)	(0.065)	(0.063)	(0.128)	(0.149)
Observations	4,157	4,157	4,157	3,944	4,153	4,153	4,153	3,940
R-squared	0.001	0.023	0.025	0.175	0.015	0.068	0.069	0.209
Year FE	YES	YES	YES	YES	YES	YES	YES	YES
Residence FE	NO	YES	YES	YES	NO	YES	YES	YES
Age group FE	NO	NO	YES	YES	NO	NO	YES	YES
Year X Age group X Residence FE	NO	NO	NO	YES	NO	NO	NO	YES
Base value	.89	.89	.89	.89	2.84	2.84	2.84	2.84

Dependent variable: *Ever had a child* (Columns 1–4), *Number of children ever had* (Columns 5–8). All estimates are based on the 1979 and 1990–1992 waves of the CIS barometer surveys. *1930/1939* is a dummy equal to 1 for women born between 1930 and 1939; *1940/1950* is a dummy for women born between 1940 and 1950. The omitted category comprises women born between 1920 and 1929. Column 1 and 5 include year fixed effects; Columns 2 and 6 add fixed effects for place of residence (provincial capital vs. rest of province); Columns 3 and 7 further include 10-year age group fixed effects; Columns 4 and 8 include the full interaction between year, residence, and age group fixed effects. Standard errors are robust to heteroskedasticity and clustered at the birth cohort-survey year level. The final row reports the mean of the dependent variable for the reference group (i.e., women born 1920–1929). *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Using the specification in Column 8 of Table 1 as reference, Figure 2 presents the corresponding

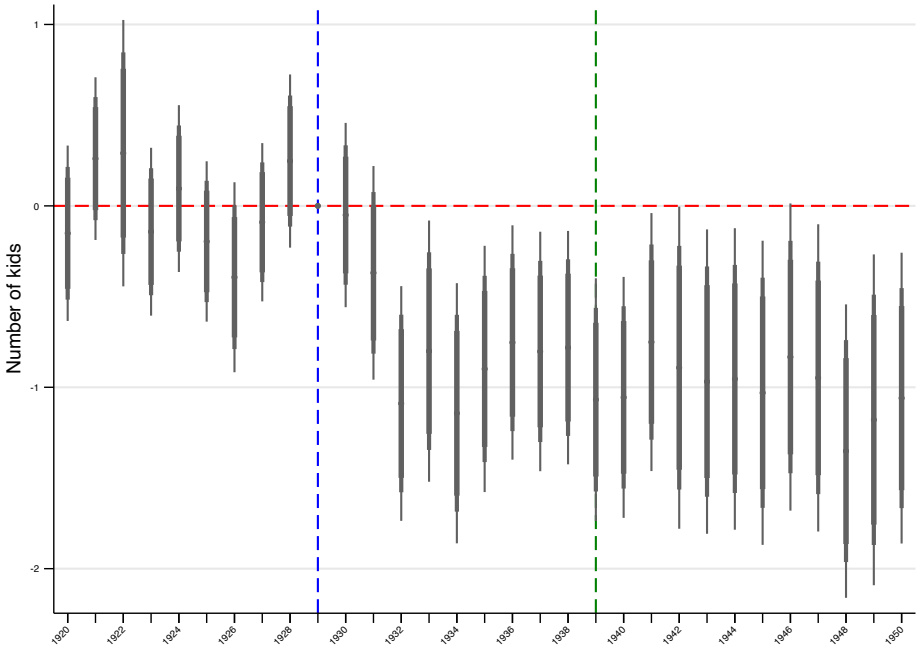
event study estimates. As shown, coefficients for the control cohorts are generally not statistically different from zero. While some estimates are noisy and fluctuate around zero, there is no evidence of a systematic pre-trend. In contrast, the negative effect of Franco's primary school reform begins to emerge with the 1932 cohort—those who were 13 years old at the time of implementation and thus spent at least three years under the reformed curriculum. When examining the post-treatment dynamics, the negative effect appears fairly stable in magnitude across cohorts, regardless of whether they were partially or fully exposed to the reform. The only notable exception is the 1930–1932 cohorts, which display a declining trend. As previously discussed in Section 4, these cohorts were only exposed to the reform during the non-compulsory years of schooling (ages 13–15). As a result, only a subset of women in these cohorts would have actually received the treatment—likely attenuating the average cohort-level effect. Moreover, this treated fraction is plausibly inversely related to the distance from the end of compulsory schooling; that is, it is likely higher for those who were 13 years old in 1945 and lower for those who were already 15, for whom remaining in school would have been less common. Taken together, these results suggest that the reform significantly reduced the total number of children women had, even among partially treated cohorts, once we account for the muted or voluntary exposure faced by those born between 1930 and 1932.

5.1 Robustness

To assess the robustness of our findings, we perform two additional exercises beyond the baseline analysis. First, we run a validation test that leverages an alternative and substantially larger dataset—the Spanish population census microdata collected by INE. In each of these cases, we re-estimate Equation 1 using the corresponding sample. Second, we run a placebo test by exploiting a sample composed of individuals from Portugal, constructed using data from the European Social Survey (ESS).

Table B3 reports the estimates from Equation 1 using the sample constructed from individual-level Spanish census data. Consistent with the baseline results presented in Table 1, we find no detectable effect at the extensive margin (i.e., the probability of ever having had a child), while a statistically significant effect emerges when the intensive margin is also considered (i.e., the total number of children). Relative to the baseline analysis, however, two key differences stand out. First, the estimates are now statistically significant for both partially and fully treated cohorts. Second, the magnitude of the effect is considerably smaller in both absolute terms and relative to the mean of the control group. Both differences likely reflect the substantial increase in sample size afforded by the census data, which improves the precision of the estimates. Quantitatively, women in partially treated cohorts have on average 0.04 fewer children compared to the control group, corresponding to approximately a 1% reduction relative to the average number of children among women born between 1920 and 1929. Women in fully treated cohorts

Figure 2: The impact of Francoist educational reform on childbearing decision: Event Study



Note: The figure shows the point estimates and the confidence intervals at 90% (i.e. widest spikes), at 95% (i.e. least-wide spikes) for each birth cohort of women in Equation 2 and comparable to the DiD estimates presented in column 8 of Table 1. The dependent variable is women' number of kids ever had. The blue dashed line indicates the reference birth cohort (i.e., the last that completed the primary school before the Franco's reform, which are women born in 1929), while the green dashed line indicates the first birth cohort fully affected by the reform, (i.e., the first that studied the entire primary school under the reform, which are women born in 1939).

have 0.08 fewer children on average, or about a 3% decline relative to the control group mean.

The analysis of the coefficients obtained by estimating Equation 1 on the sample of Portuguese women from the ESS are presented in Table B4. Their inspection reassures us that our findings are not merely driven by generational dynamics. Despite Portugal being highly comparable to Spain under Franco—in terms of geographic proximity, strong Catholic tradition, and the presence of an authoritarian regime—women in the same birth cohorts that were partially or fully exposed to Franco's reform (in temporal terms) but born in Portugal show no significant changes in childbearing behavior, either at the extensive or intensive margin. This placebo result strengthens our interpretation that the effects observed in Spain are indeed attributable to exposure to the 1945 primary school reform, rather than to broader cohort-based trends affecting Southern European countries during this period.

5.2 Mechanisms

Multiple mechanisms may underlie the effects documented in Table 1. First, a labor market substitution mechanism may be at play: women who were partially or fully treated by the reform may have increased their participation in the labor force, thereby reducing the time available for childbearing—particularly when it comes to having more than one child. Conversely, the reform may have reduced the human capital of the treated cohorts by shifting the curriculum towards domesticity and motherhood-related instruction at the expense of skills that are directly valuable in the labor market. This reduction in labor market prospects may have, in turn, generated frustration or a sense of disempowerment, which could have contributed to lower fertility preferences or outcomes. Complementary to these labor market channels, another plausible mechanism may operate through political identity. Being exposed to an authoritarian, far-right regime that promoted traditional family values through education may have triggered a long-term ideological backlash among affected women, fostering more progressive or left-leaning attitudes that rejected the pro-natalist, conservative ideals associated with the regime. Finally, the reform may have directly altered both the primary and secondary beliefs of treated women regarding motherhood, gender roles, and the place of women in society, potentially affecting their attitudes toward childbearing more generally.

To explore the mechanisms that may drive our baseline findings, we re-estimate the specification in Equation 1 using a set of alternative dependent variables: labor force participation, highest level of educational attainment, political orientation (measured as whether the respondent voted for right-wing parties in the most recent elections), and a battery of survey items capturing both primary and secondary beliefs related to motherhood. These belief-based outcomes include views on women's role within the family, attitudes toward children, labor market participation,

and alignment with Catholic Church teachings on fertility-related matters.[20]

As reported in Table B5, we do not detect any statistically significant differences across untreated, partially treated, and fully treated cohorts with respect to labor market participation, tertiary educational attainment, or voting behavior in favor of right-wing parties. These null results allow us to rule out most of the previously discussed channels as primary drivers of our main findings.

By contrast, Figure 3 offers graphical evidence on the estimated coefficients from DiD regressions using individual answers to survey questions on beliefs about women's roles in the family, the labor market, and motherhood more broadly. Strikingly, all estimated effects are directionally consistent with a mechanism whereby exposure to the reform directly shaped treated women's values and beliefs, potentially inducing long-lasting shifts in cultural attitudes. In particular, fully treated cohorts are more likely than the control group to support gender equality in the domestic sphere, to reject the notion that men should do less housework, and to view equal task-sharing as essential for couple stability. These women also attribute lower importance to children in defining female identity or relationship stability, and place greater emphasis on women's participation in the labor market as a key component of family well-being. Finally, fully treated women are systematically less aligned with the Catholic Church's traditional positions on fertility-related issues, including abortion, artificial insemination, and contraception. Partially treated cohorts exhibit similar patterns, although the magnitude of the estimated effects is generally smaller and some coefficients fall short of conventional significance thresholds.

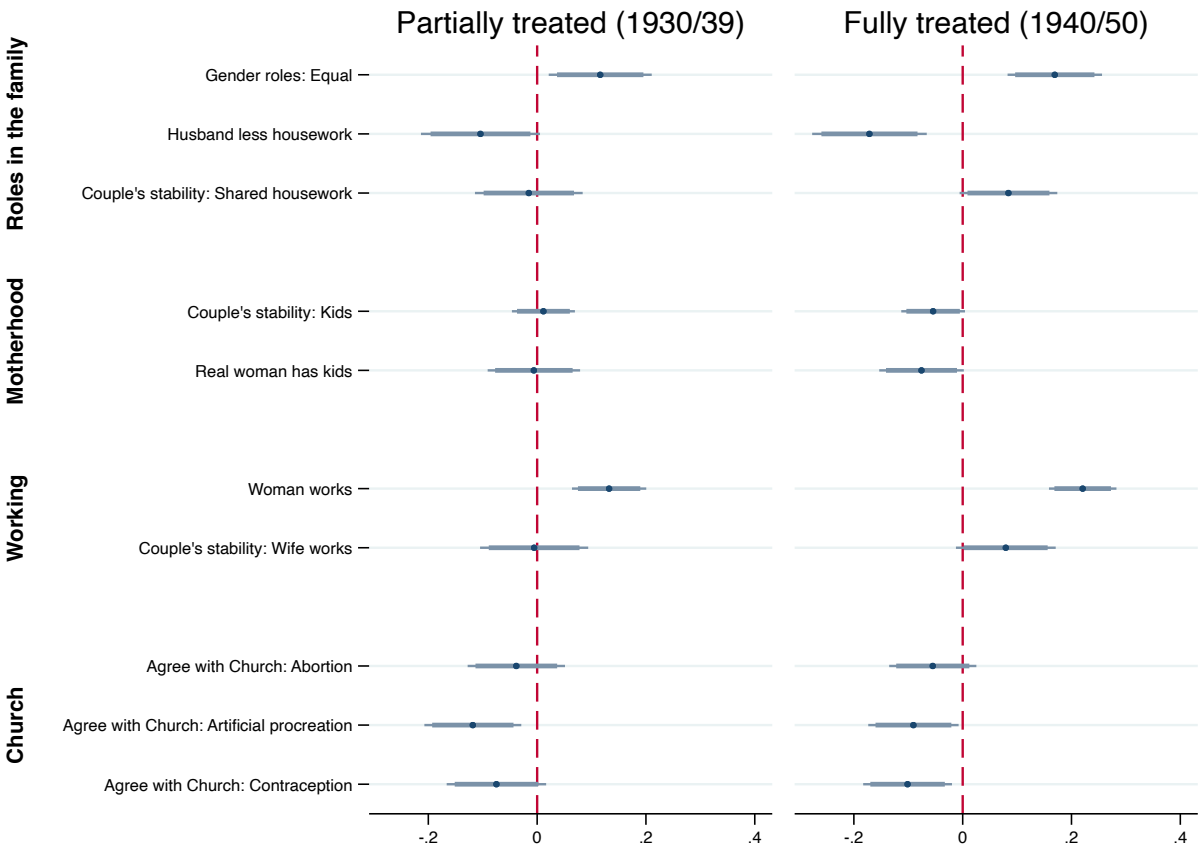
While these findings should be interpreted with caution given the data limitations discussed earlier, they offer at least strongly suggestive evidence in support of a cultural backlash mechanism. Specifically, our results are consistent with the hypothesis that exposure to the Francoist primary school reform—an intervention explicitly aimed at reinforcing traditional gender norms—may have induced a lasting rejection of the very same values it sought to impose, particularly among women most intensely exposed to it.

5.3 Impact at the macro level

Finally, we investigate whether the individual-level behavioral changes induced by the reform translated into detectable aggregate effects—specifically, whether fertility declined more sharply in areas where the population was more intensely exposed to the policy. To test this, we adopt a difference-in-differences (DiD) approach at the aggregate level (provincial capitals and the rest of the provinces) and compare fertility trends before and after the 1945 primary school

[20]Belief-related variables are only available in a single survey wave (1990), which precludes the inclusion of year or age-class fixed effects in the corresponding regressions. Similarly, standard errors cannot be clustered at the year-by-year-of-birth level and are instead adjusted for heteroskedasticity. As a result, this set of estimates should be interpreted as suggestive rather than causal.

Figure 3: The impact of Francoist educational reform on personal beliefs: DiD



Note: The figure shows the point estimates and the confidence intervals at 90% (i.e. widest spikes), at 95% (i.e. least-wide spikes), of the exposure to Franco's educational reform on personal beliefs, for women born between 1930 and 1939 (left panel) and women born between 1940 and 1950 (right panel), both with respect to women born between 1920 and 1929. The dependent variables are women' beliefs on gender roles within the household, motherhood, women' participation to the labor force and allignement with Catholic values. All regressions include place of residence fixed effects and adjust standard errors for heteroskedasticity.

reform across areas with differential exposure. In particular, we contrast locations with a higher pre-reform share of girls aged 0–14 in 1940—i.e., the cohorts who would be the first to be (partially or fully) affected by the policy—with those where this share was lower. We estimate the following equation:

$$y_{it} = \beta_0 + \beta_1 \, Treat_i \times Post_t + \mathbf{X}_{it} + \Gamma_i + \tau_{pt} + \epsilon_{it} \tag{3}$$

where i denotes an administrative unit—either a provincial capital or the remainder of the province—and t indicates the year. The outcome variable y_{it} is the number of live births in unit i in year t. The interaction term $Treat_i \times Post_t$ combines $Post_t$, a dummy equal to 1 for years after 1945 (i.e., the implementation of the Primary Education Law), and $Treat_i$, which captures the share of the female population aged 0–14 in 1940—corresponding to the cohorts who would have been of school age at the time of the reform. We also include a set of controls $\mathbf{X}_{it}$, which comprises the interaction between $Post_t$ and the share of the male population aged 0–14 in 1940, as well as the number of women of reproductive age in each unit and decade, to ensure comparability of raw birth counts across space and time, as discussed in Section 3. The specification further includes unit fixed effects (Γ_i), which absorb time-invariant characteristics at the local level, and province-by-year fixed effects (τ_{pt}), which account for time-varying shocks that may differentially affect Spanish provinces (e.g. the effect of the Spanish civil war). Standard errors are robust to heteroskedasticity and clustered at the level of the administrative unit (i). As with the baseline results, we also estimate an event-study version of Equation 3 to assess the validity of the parallel trends assumption and to gain additional insights into the dynamics of the post-treatment effects. Table B6 reports balancing tests between treated and control areas across a range of local characteristics, including proxies for religiosity, demographic composition of the resident population, political orientation of the electorate, and features of the local educational system.[21] In virtually all cases, we find no statistically significant differences between more and less exposed areas to the reform.[22]

Table 2 reports the estimated coefficients from Equation 3. Columns 1 and 3 include year and unit fixed effects, as well as controls for the demographic composition of the fertile female population by age group and decade. Columns 2 and 4 further account for province-by-year

[21]For these tests, we estimate cross-sectional regressions that mirror as closely as possible Equation 3, changing only the dependent variable. As these are cross-sectional regressions, we control for province fixed effects and for a dummy distinguishing provincial capitals to the rest of the province.

[22]The only exception is the share of votes received by the Socialist party in the last free elections held in 1931, which is slightly lower in areas more exposed to Franco's educational reform. That is, these areas appear to have historically had weaker support for socialist parties. While the only requirement for the identification strategy is the validity of the parallel trends assumption, we nonetheless run robustness tests that allow for non-parametric differential trends based on pre-treatment socialist vote shares. These controls do not alter the results in terms of either magnitude or significance.

fixed effects, thereby absorbing time-varying shocks at the provincial level. In addition, Columns 3 and 4 incorporate an interaction term between the share of males aged 0-14 in 1940 and the post-treatment indicator (*Post*). Across all specifications, the coefficient on our main treatment variable—the interaction between the share of women aged 0-14 in 1940 and *Post*—is negative and statistically significant.[23] According to the estimate in the fully saturated specification (Column 4), a one-percentage-point increase in the share of the female population exposed to the primary school reform leads to a decrease of approximately 460 live births per year. This corresponds to roughly 9% of the average number of births recorded in the statistical units over the five years preceding the introduction of the reform (i.e., 1939–1944). As a placebo, it is worth noting that the coefficient on the analogous interaction term for males (i.e., share of men aged 0-14 in 1940 × *Post*) is consistently much smaller—roughly one-third the size of the female coefficient—and never statistically significant. This pattern is consistent with the interpretation that the reform targeted girls directly and had limited spillover effects on male cohorts.

Table 2: The impact of Francoist educational reform on the number of births: DiD

	(1) Births	(2) Births	(3) Births	(4) Births
Treat (W) X Post 1945	-429.396***	-519.120**	-392.784***	-459.933*
	(115.321)	(200.829)	(146.955)	(234.202)
Treat (M) X Post 1945			-55.138	-163.600
			(86.689)	(138.879)
Observations	4,919	4,918	4,919	4,918
R-squared	0.795	0.891	0.795	0.892
Unit FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
Prov. X Year FE	NO	YES	NO	YES
Mean of depvar	5183	5183	5183	5183

Dependent variable: *Number of births*. The variable *Treat (W)* × *Post* represents the interaction between the share of women aged 0–14 in 1940 in each observational unit (*Treat (W)*) and the post-treatment indicator *Post*, which equals 1 for years after 1945 and 0 otherwise. Similarly, *Treat (M)* × *Post* is the interaction between the share of men aged 0–14 in 1940 (*Treat (M)*) and *Post*. All specifications include unit and year fixed effects, as well as controls for the size of the female reproductive-age population by age group and decade. Columns 2 and 4 additionally include province-year fixed effects. Standard errors are heteroskedasticity-robust and clustered at the unit level. The last row reports the average number of live births per unit in the pre-treatment period (1939–1944). *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Building on the specification reported in Column 4 of Table 2, Figure 4 presents the corresponding event-study analysis. As shown, we detect no statistically significant differences in the number of live births between treated and control areas in the years prior to the introduction of the reform, lending support to the parallel trends assumption. By contrast, statistically significant

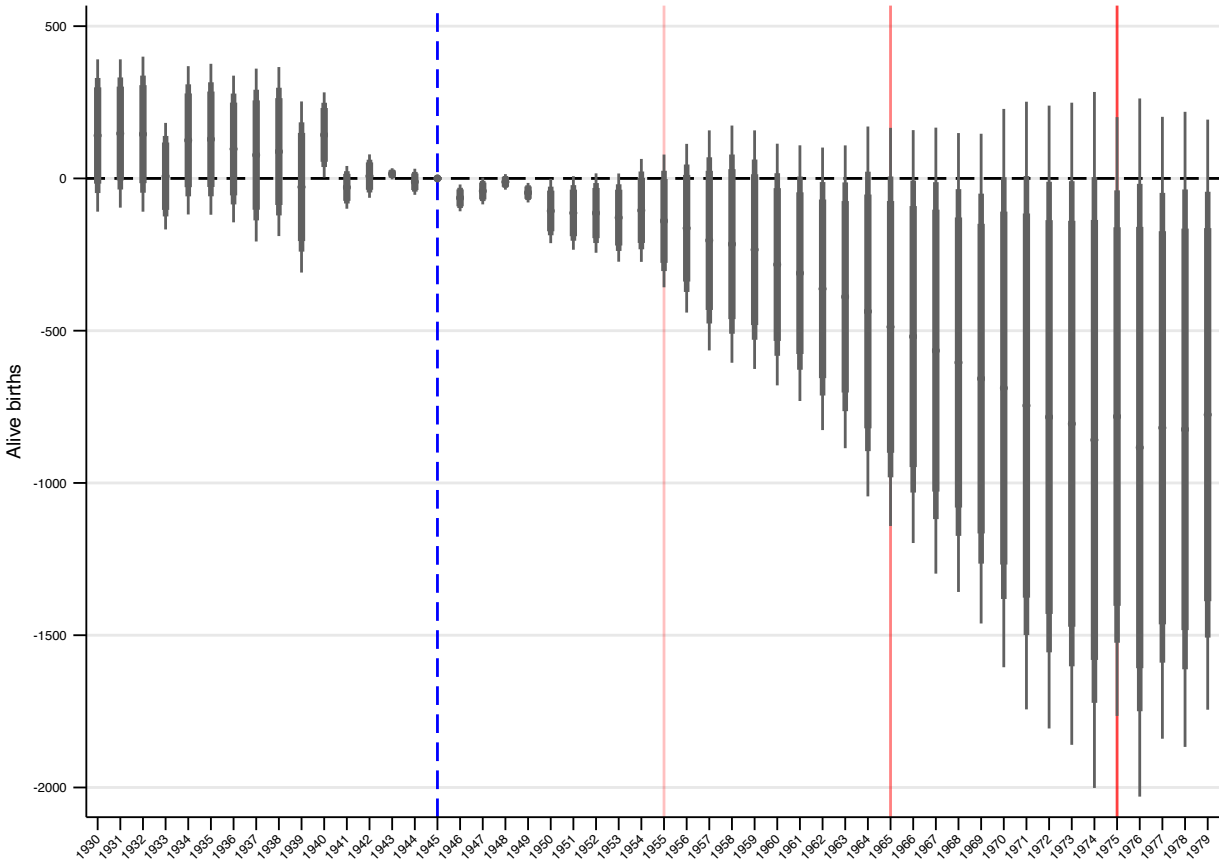
[23]The effect becomes slightly noisier in Column 4 when both province-year fixed effects and the male-share interaction are included; the p-value in this case is 0.052.

and increasingly negative differences emerge shortly after the reform, growing in magnitude over time. This dynamic is mechanically driven by changes in the composition of the fertile female population: as time progresses, the cohorts exposed to the reform—either partially or fully—gradually replace older, unexposed cohorts, thereby increasing the share of women subject to the reform within the childbearing age range. Vertical red lines in Figure 4 help mark this transition. The lightest red line indicates the year when the youngest fertile age group (15-24) becomes entirely composed of partially exposed women; the medium-red line corresponds to the point when the 15-24 group is fully treated and the 25–34 group is composed entirely of partially treated women; and finally, the darkest red line marks the year when the entire fertile population (ages 15-44) is composed exclusively of women either partially or fully exposed to the reform. Consistent with our expectations, the estimated treatment effect peaks around the latter threshold and stabilizes thereafter at a significantly negative level.

Table B7 presents the results of four different robustness checks. In Columns 1 and 2, we address the potential imbalance in the share of votes for the Socialist Party in the 1931 elections—documented in Table B6—by including non-parametric trends based on this variable. Columns 3 and 4 replace the continuous treatment measure with a discretized version of this measure, as here we compare areas above and below the median of the exposure distribution. In Columns 5 and 6, we perform a placebo test using as treatment the share of women aged 15-24 in 1940—a group that could not have been affected by the reform. Finally, Columns 7 and 8 investigate whether the reform might have indirectly influenced birth rates through changes in marriage rates, given that marriage was at the time a near-prerequisite for childbearing decisions. All tests yield results that are consistent with our baseline findings and support the robustness of the main estimates presented in Table 2. Specifically, controlling for pre-existing political alignment (i.e., Socialist vote share in 1931) has virtually no impact on the estimated effects. The discrete version of the treatment produces results that closely mirror those from the continuous specification, both in sign and statistical significance. The magnitude of the estimated effect is substantial: being above rather than below the median exposure reduces the number of live births by approximately 5,000, or about 100% of the pre-reform average. As expected—and consistent with results using male exposure as placebo—replacing the treatment variable with the share of women aged 15-24 in 1940 yields a coefficient that is statistically indistinguishable from zero. The same holds when we examine marriage rates: areas with higher concentrations of reform-exposed women (aged 0-14 in 1940) do not exhibit significant differences in marriage outcomes relative to less-exposed areas.

Finally, we seek to rule out the possibility that the estimated effects reported in Table 2 simply reflect the broader process of the second demographic transition—a common phenomenon in industrializing countries. As discussed in Section 3, due to data limitations, we are unable to construct suitable fertility measures for local units (e.g., Total Fertility Rate, TFR) that vary over

Figure 4: The impact of Francoist educational reform on childbearing decisions: Event Study



Note: The figure shows the point estimates and the confidence intervals at 90% (i.e. widest spikes), at 95% (i.e. least-wide spikes) for each birth cohort of women of the event-study version of Equation 3 and comparable to the DiD estimates presented in column 4 of Table 2. The dependent variable is the number of live births in unit i at time t. The lightest red line indicates the year when the youngest fertile age group (15-24) becomes entirely composed of partially exposed women; the medium-red line corresponds to the point when the 15-24 group is fully treated and the 25–34 group is composed entirely of partially treated women. The darkest red line marks the year when the entire fertile population (ages 15-44) is composed exclusively of women either partially or fully exposed to the reform

time and allow for spatial and temporal comparisons before and after Spain's industrialization. Nevertheless, in Figure B2, we report national-level TFR estimates by decade, from 1850 to 2020, as reconstructed by the Spanish Statistical Institute (INEE). While the figure confirms a long-run declining trend in fertility, it also shows that the TFR remained relatively stable during the post-treatment decades relevant to our analysis (i.e., 1940s–1970s). This comparison provides suggestive evidence that the observed decline in fertility is unlikely to be driven solely by the demographic transition and instead reflects a genuine effect of the Franco-era education reform.

6 Conclusion

This paper examines the demographic consequences of the 1945 Francoist primary education reform in Spain, which sought to increase fertility by reinforcing traditional gender roles and promoting a nationalist-religious vision of women as wives and mothers. Exploiting quasi-exogenous variation in exposure across birth cohorts and localities within a difference-in-differences framework, we show that the reform not only failed to achieve its intended goals but in fact reduced fertility. Women who completed their primary education under the new curriculum bore significantly fewer children than comparable cohorts, with the effect operating primarily at the intensive margin. Complementary evidence suggests that this fertility decline was accompanied by a cultural backlash: treated women were more likely to reject the regime's promoted values around gender roles, family, and motherhood. At the aggregate level, areas with a larger share of girls exposed to the reform subsequently experienced a marked reduction in fertility rates. Taken together, even though data limitations prevent us from isolating the contribution of each component of the reform—such as differences in curricula, class division, or teaching materials for boys and girls—, these results demonstrate that attempts to engineer social norms through indoctrination in school may yield outcomes directly opposite to those envisioned by policymakers.

Our findings have important implications for contemporary policy debates. In a growing number of advanced economies, low fertility has reignited calls for a "return" to traditional values as a path to demographic recovery. Public figures in the U.S. and several other Western democracies have advocated restoring more conservative family structures as a way to encourage childbearing. The evidence presented here challenges this narrative. Far from stimulating fertility, policies that seek to reassert gender stereotypes risk alienating those they target, thereby depressing rather than boosting demographic outcomes. In modern low-fertility settings, where women often bear the largest costs of childbearing, attempts to reinforce traditional gender roles are especially likely to backfire.

This study generates four lessons for pro-natalist policy. First, unlike economic incentives,

indoctrination through education is shown to be counterproductive, adding a new dimension to our understanding of policy effectiveness. Second, it provides causal evidence that state-led efforts to entrench traditional roles can produce long-lasting changes in attitudes in the opposite direction. Third, it shows that indoctrination and authoritarian propaganda have the potential to extend beyond beliefs into core demographic behavior. Finally, our results highlight the limits of authoritarian attempts to reshape intimate life.

In conclusion, the Francoist reform illustrates the demographic risks of using education as a vehicle for ideological engineering. Rather than securing a "demographic recovery," the policy provoked a cultural and demographic backlash with enduring effects. For today's policymakers confronting low fertility, the lesson that effective pro-natalist strategies are not those that attempt to reimpose outdated gender norms appears clear.

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Appendix

A Data

Table A1: Summary Statistics

	Mean	Std. Dev.	Min	Max	N
Panel A: CIS Data					
Kids	0.902	0.298	0.00	1.00	4,157
N. kids	2.661	1.685	0.00	16.00	4,153
Gender roles: Equal	0.507	0.500	0.00	1.00	1,101
Husband less housework	0.545	0.498	0.00	1.00	903
Couple's stability: Shared housework	0.682	0.466	0.00	1.00	1,115
Couple's stability: Kids	0.842	0.364	0.00	1.00	1,111
Real woman has kids	0.639	0.480	0.00	1.00	1,081
Woman works	0.295	0.456	0.00	1.00	1,130
Couple's stability: Wife works	0.494	0.500	0.00	1.00	1,073
Agree with Church: Abortion	0.333	0.471	0.00	1.00	1,343
Agree with Church: Artificial procreation	0.267	0.443	0.00	1.00	1,204
Agree with Church: Contraception	0.266	0.442	0.00	1.00	1,198
Panel B: Census Microdata					
Kids	0.916	0.278	0.00	1.00	1,140,219
N. kids	2.649	1.690	0.00	20.00	1,140,219
Panel C: Macro level data					
Births	6001.111	7349.494	11.00	92270.00	4,919

Table A2: Questions from CIS

Survey *(Numb.)*	**Year**	**Question**
		Roles in the family
Intra-household inequality *(1867)*	1990	*Gender roles: equal.* As you know, there are currently different types of families. Of the three options I am about to read to you, could you tell me which one best matches your ideal family? ***a)*** A family in which both the man and the woman work outside the home and share household chores and childcare. *b)* A family in which the woman works fewer hours outside the home and therefore takes on a greater share of household chores and childcare. *c)* A family in which only the man works outside the home, and the woman is exclusively responsible for household chores and childcare.

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Table A2 – continued from previous page

Survey *(Number)*	**Year**	**Question**
Intra-household inequality *(1867)*	1990	*Husband less housework.* To what extent do you think a man should share household chores? *a)* Equally with the woman. *b)* To a lesser extent than the woman. *c)* Only in case of necessity.
Intra-household inequality *(1867)*	1990	*Couple's stability: Shared housework.* Now I am going to read you a series of attitudes and situations that some people believe contribute to a happy marriage. For each one, please tell me whether you think it is very important, quite important, slightly important, or not important at all for achieving the stability of the couple: sharing household chores.
		Motherhood
Intra-household inequality *(1867)*	1990	*Couple's stability: Kids.* Now I am going to read you a series of attitudes and situations that some people believe contribute to a happy marriage. For each one, please tell me whether you think it is very important, quite important, slightly important, or not important at all for achieving the stability of the couple: having children.
Intra-household inequality *(1867)*	1990	*Real woman has kids.* In the case of women, do you believe that being a mother is an experience one must have in order to truly feel like a woman? *a)* Yes, *b)* No, *c)* Don't know, *d)* No answer.
		Working
Intra-household inequality *(1867)*	1990	*Woman works.* Could you tell me if you currently have paid employment? Yes, No, No answer.
Intra-household inequality *(1867)*	1990	*Couple's stability: Wife works.* Now I am going to read you a series of attitudes and situations that some people believe contribute to a happy marriage. For each one, please tell me whether you think it is very important, quite important, slightly important, or not important at all for achieving the stability of the couple: the woman works outside the home.
		Church
February Barometer*(1860)*	1990	*Agree with Church: abortion.* Do you tend to agree or disagree with the recommendations made by the Catholic Church regarding abortion?
February Barometer *(1860)*	1990	*Agree with Church: artificial procreation.* Do you tend to agree or disagree with the recommendations made by the Catholic Church regarding artificial fertilization methods?

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Table A2 – continued from previous page

Survey *(Number)*	**Year**	**Question**
February Barometer *(1860)*	1990	*Agree with Church: contraception.* Do you tend to agree or disagree with the recommendations made by the Catholic Church regarding contraceptives?

B Additional Tables and Figures

Table B3: The impact of Francoist educational reform on childbearing decision: DiD - Census

	(1) Kids	(2) N.Kids
1930/1939	-0.005	-0.035***
	(0.008)	(0.012)
1940/1950	-0.006	-0.078***
	(0.010)	(0.022)
Observations	1,140,203	1,140,203
R-squared	0.008	0.027
Year FE	YES	YES
Residence FE	YES	YES
Age class FE	YES	YES
Base value	.91	2.78

Dependent variable: *Ever had a child* (Column 1), *Number of children ever had* (Column 2). All estimates are based on the 1991 and 2011 waves of the Spanish Census. *1930/1939* is a dummy equal to 1 for women born between 1930 and 1939; *1940/1950* is a dummy equal to 1 for those born between 1940 and 1950. The omitted reference category includes women born between 1920 and 1929. All specifications include the full interaction between year, municipality of residence, and age group fixed effects. Standard errors are robust to heteroskedasticity and clustered at the birth cohort × survey year level. Standard errors are robust to heteroskedasticity and clustered at the birth cohort × survey year level. The last row reports the mean of the dependent variable for the reference category (i.e., women born between 1920 and 1929). *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Table B4: The impact of Francoist educational reform on childbearing decision: DiD - Portugal

	(1) Kids	(2) N. kids
1930/1939	-0.019	-0.629
	(0.020)	(0.399)
1940/1950	-0.009	-0.484
	(0.038)	(0.429)
Observations	751	751
R-squared	0.015	0.026
Year FE	YES	YES
Age class FE	YES	YES
Base value	.88	2.89

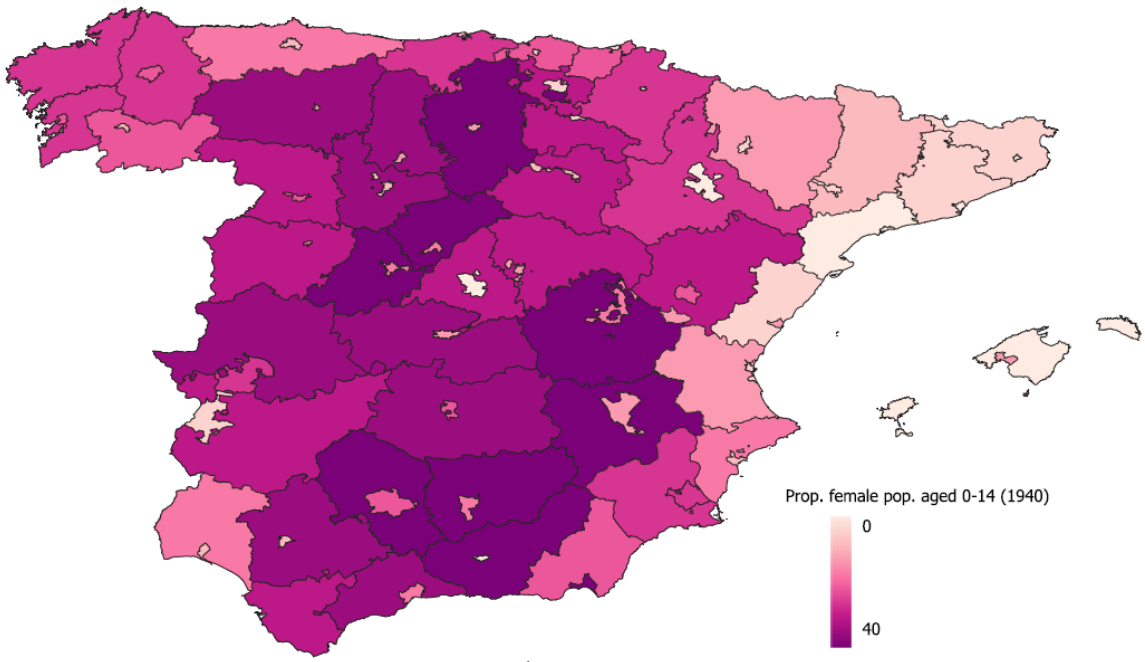
Dependent variable: *Ever had a child* (Column 1), *Number of children ever had* (Column 2). All estimates are based on the 2006 and 2018 waves of the European Social Survey (ESS) for Portugal. *1930/1939* is a dummy equal to 1 for women born between 1930 and 1939; *1940/1950* is a dummy equal to 1 for those born between 1940 and 1950. The omitted reference category comprises women born between 1920 and 1929. All specifications include year-age group fixed effects. Standard errors are robust to heteroskedasticity and clustered at the birth cohort × survey year level. The final row reports the mean of the dependent variable for the reference group (i.e., women born 1920–1929). *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table B5: The impact of Francoist educational reform on labor force partecipation, education and political ideology: DiD

	(1) Employed	(2) Unemployed	(3) Housewife	(4) Educated	(5) Right-wing
1930/1939	0.017	0.006	-0.024	-0.004	-0.069
	(0.052)	(0.011)	(0.050)	(0.015)	(0.057)
1940/1950	0.075	-0.001	-0.074	0.002	0.032
	(0.058)	(0.018)	(0.058)	(0.020)	(0.097)
Observations	3,241	3,241	3,241	3,490	1,033
R-squared	0.253	0.190	0.257	0.162	0.327
Year FE	YES	YES	YES	YES	YES
Residence FE	YES	YES	YES	YES	YES
Age class FE	YES	YES	YES	YES	YES
Base value	.27	.03	.69	.02	.3

Dependent variables: *Employed (col. 1), Unemployed (col. 2), Housewife (col. 3), Tertiary Education (col. 4), Voted for Right-Wing Parties (col. 5).* All results except column 5 rely on the 1979 and 1986–1988 waves of the CIS Barometer; column 5 uses only the 1986–1988 waves. *1930/1939* is a dummy equal to 1 for women born between 1930 and 1939; *1940/1950* is a dummy for women born between 1940 and 1950. The reference group consists of women born between 1920 and 1929. All specifications include the full set of fixed effects defined by the interaction between year, place of residence, and age group. Standard errors are robust to heteroskedasticity and clustered at the year-by-year-of-birth level. The last row reports the average value of the dependent variable for the reference group (i.e., women born between 1920 and 1929). *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Figure B1: Proportion of female population aged 0-14 in 1940



Note: The map shows the proportion of the female population aged 0–14 in 1940. Darker shades indicate a higher proportion of young females relative to the total female population.

Table B6: Treated vs Control - Balancing Tests

	(1) Coefficient	(2) SD
Purged teachers	0.328	0.818
Religious communities (1930)	-0.006	0.008
Professed (pc, 1930)	-0.175	0.125
Novices (pc,1930)	-0.020	0.051
Lay (pc, 1930)	-0.020	0.028
Republicans (%, local 1931)	0.465	0.487
Socialists (%, local 1931)	-0.567**	0.274
Monarchic (%, local 1931)	0.015	0.537
Comunsits (%, local 1931)	0.014	0.031
Pro regime (%, 1947)	0.242	0.410
National schools (1000s,1942)	0.019	0.031
Primary teachers (1000s,1942)	0.012	0.030
Enrolled students (1000s,1942)	-0.040	1.239
Area in the war front (%)	0.234	0.738
Women aged 15-24 (%, 1940)	0.000	0.002
Women aged 25-34 (%, 1940)	0.000	0.001
Women aged 35-44 (%, 1940)	-0.000	0.001
Women aged 15-24 (1940)	-781	608
Women aged 25-34 (1940)	-764	553
Women aged 35-44 (1940)	-668	415
Male deaths (1000s, 1936-39)	0.289	0.714
Female deaths (1000s, 1936-39)	0.029	0.409

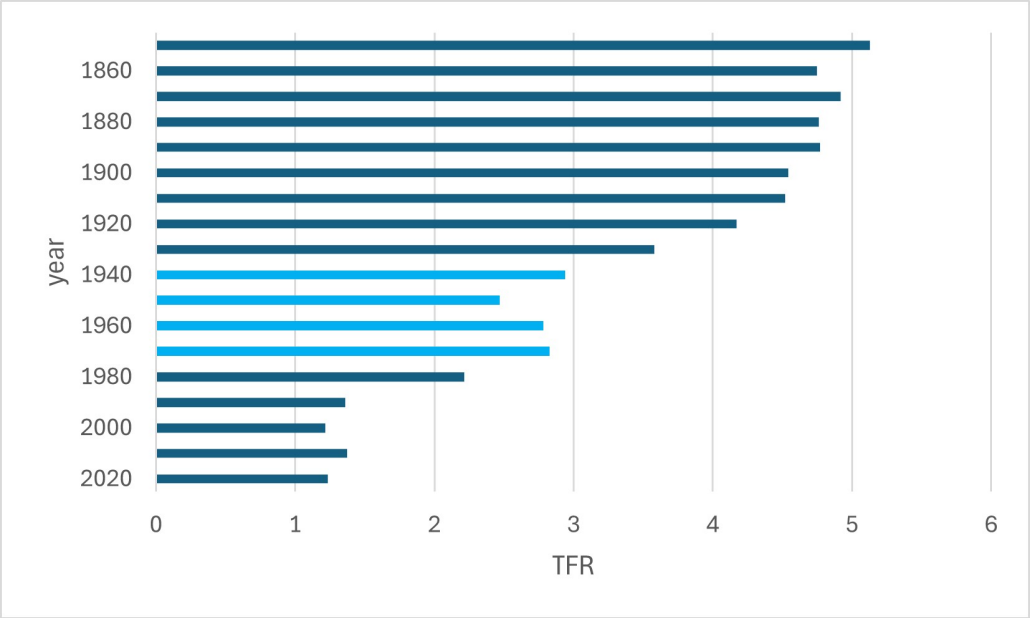
Note: This table presents balancing tests between treated and control areas across various local characteristics. We estimate cross-sectional regressions that closely follow the structure of Equation 3, changing only the dependent variable. All specifications include province fixed effects and a dummy variable distinguishing provincial capitals to the rest of the province.

Table B7: The impact of Francoist educational reform on the number of births: DiD - Robustness

	(1) Births	(2) Births	(3) Births	(4) Births	(5) Births	(6) Births	(7) Weddings	(8) Weddings
Treat (W) X Post 1945	-384.141***	-444.410**	-4,915.460***	-5,103.000**	122.977	107.961	-40.537	-62.790
	(138.237)	(214.473)	(1,701.797)	(2,417.523)	(115.894)	(111.838)	(55.894)	(65.155)
Treat (M) X Post 1945		-167.086		324.529		42.671		61.509
		(169.578)		(1,456.580)		(49.064)		(41.525)
Observations	4,919	4,918	4,918	4,918	4,918	4,918	4,918	4,918
R-squared	0.795	0.903	0.891	0.891	0.891	0.891	0.911	0.912
Test	Socialists	Socialists	Discrete	Discrete	Placebo	Placebo	Weddings	Weddings
Fertile pop.	YES	YES	YES	YES	YES	YES	YES	YES
Unit FE	YES	YES	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES	YES	YES
Prov. X Year FE	YES	YES	YES	YES	YES	YES	YES	YES
Mean of depvar	5183	5183	5183	5183	5183	5183	1728	1728

Dependent variable: *Number of births (cols. 1–6); Number of weddings (cols. 7–8).* In columns 1–2 and 7–8, *Treat (w)* × *Post* denotes the interaction between the share of women aged 0–14 in 1940 in each observational unit (*Treat (w)*) and the post-treatment indicator *Post*, which equals 1 for years after 1945 and 0 otherwise. In columns 3–4, *Treat (w)* is a dummy equal to 1 if the unit belongs to the upper half of the distribution of the share of women aged 0–14 in 1940, and 0 otherwise. In columns 5–6, *Treat (w)* refers to the share of women aged 15–24 in 1940. All specifications include unit and year fixed effects, as well as controls for the size of the female reproductive-age population by age group and decade. Even-numbered columns additionally include province-year fixed effects. Columns 1–2 also control for the interaction between year fixed effects and the share of votes received by the Socialist party in 1931 at the unit level. Standard errors are robust to heteroskedasticity and clustered at the unit level. The last row reports the average number of births per unit during the pre-treatment period (1939–1944). *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Figure B2: Total Fertility Rate in Spain



Note: The figure displays the Total Fertility Rate (TFR) in Spain from 1860 to 2020. TFR represents the average number of children a woman would have over her lifetime based on age-specific fertility rates observed in a given year. Sources: Instituto Nacional de Estadística (INE), United Nations Population Division, and Statista.